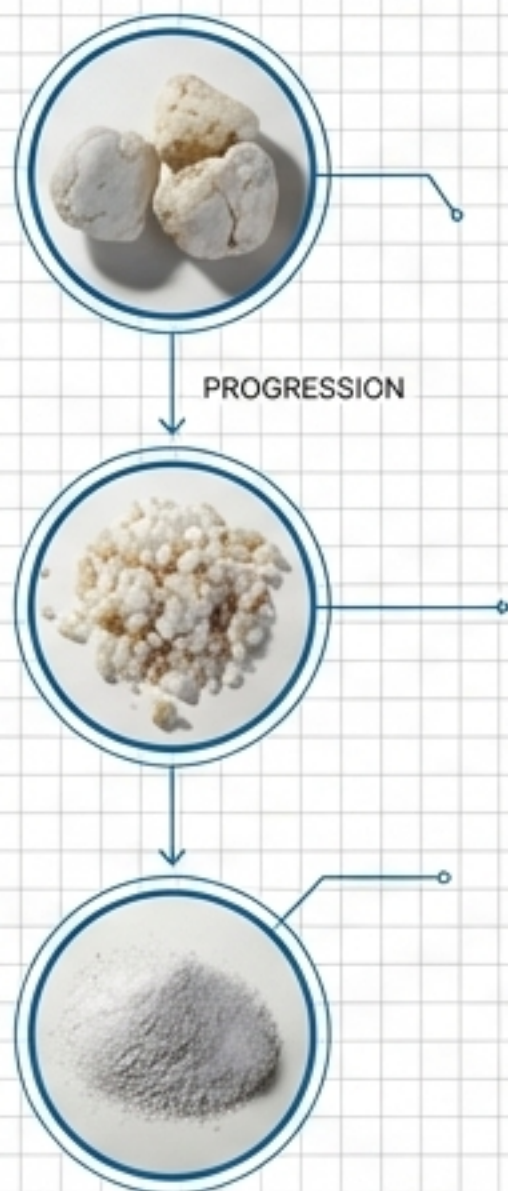
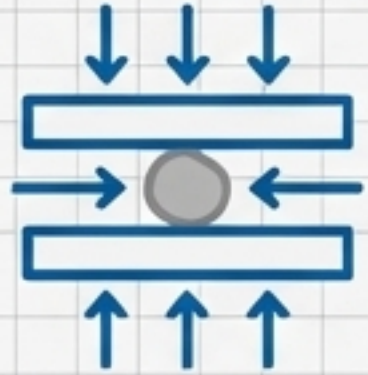




The Expert of Milling to Nano-scale

FRITSCH Laboratory Solutions Architecture

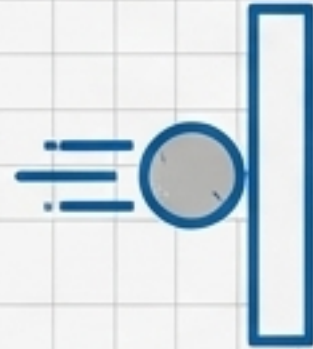
Diagnostic Matrix: Comminution Physics



Pressure

Ceramics
& Minerals

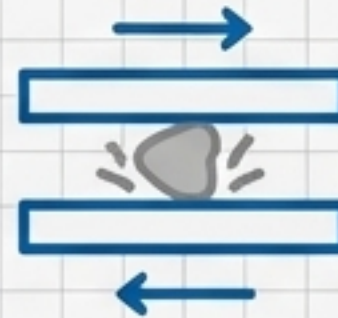
→ **Jaw Crusher**



Impact

Coal, Ores
& Alloys

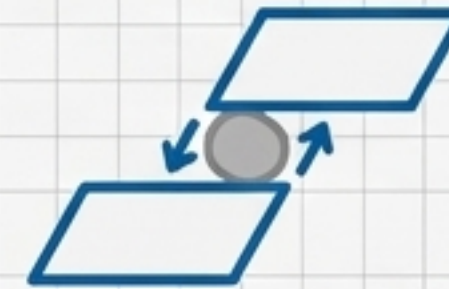
→ **Planetary &
Ball Mills**



Friction

Sediments
& Slags

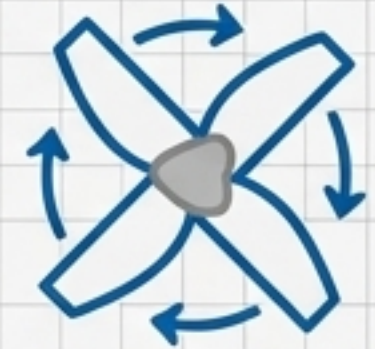
→ **Disk Mill &
Mortar**



Shearing

Building
Materials

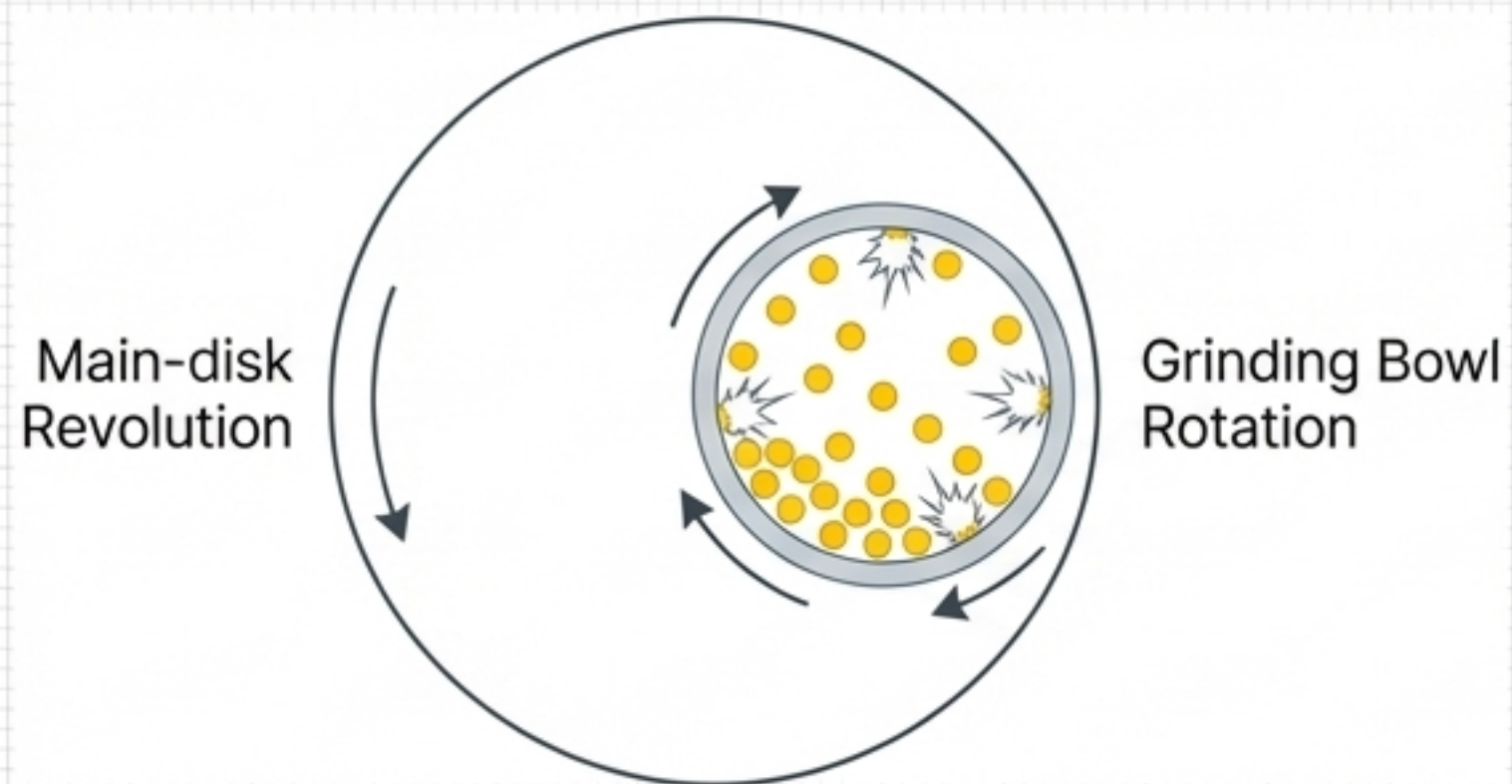
→ **Rotor Mill**



Cutting

Plastics &
Electronic Scrap

→ **Cutting Mill**



Flagship Kinematics: Planetary Ball Mills

SYSTEM KINEMATICS

Powerful impact energy generated by opposing rotational forces.

PEAK TELEMETRY

95G

95G Max Centrifugal Acceleration.

TARGET FINENESS

<0.1 μm (Wet Milling).

ENVIRONMENTAL

Inert Gas (N₂, Ar) Compatible.



Architecture Bifurcation: Classic vs. Premium

CLASSIC ARRAY



Planetary Ball Mill
PULVERISETTE 5



Planetary Mono Mill
PULVERISETTE 6



Planetary Micro Mill
PULVERISETTE 7



Vario-Planetary Mill
PULVERISETTE 4

ENERGY OUTPUT:	Up to 50G
ROTOR SPEED:	Max 800 rpm
EFFICIENCY:	Standard <1 μm

PREMIUM ARRAY



Planetary Mill
PULVERISETTE 5 Premium



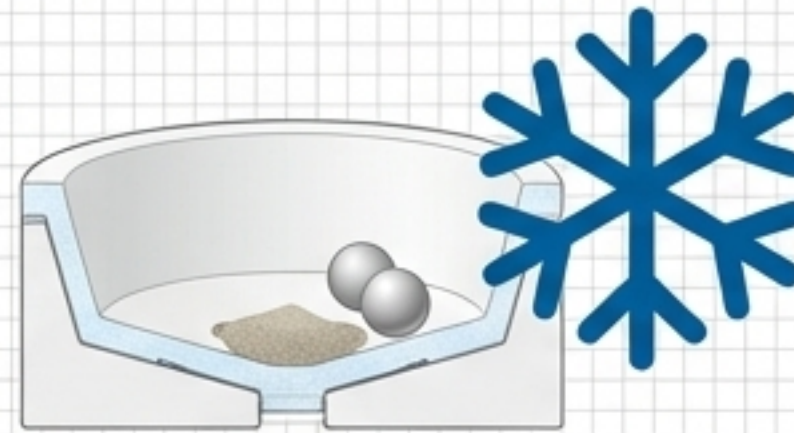
Planetary Micro Mill
PULVERISETTE 7 Premium

ENERGY OUTPUT:	Up to 95G (3x Standard Energy)
ROTOR SPEED:	Max 1,100 rpm
EFFICIENCY:	Extreme Nano-scale <0.1 μm

PREMIUM UPGRADES DETECTED: ServoLOCK & SelfLOCK bowl clamping, Automatic Check validation, Minimum 0.05mm ball compatibility.

Micro-Scale & Cryogenic Protocols

PULVERISETTE 23: Vibratory Mini Mill



Cryogenic PTFE Bowl Architecture

SYSTEM:

Vibratory Mini Mill (P23)

VOLUME PROTOCOL:

Sub-3ml Micro-Processing

THERMAL STATUS:

Liquid Nitrogen freeze-milling supported via PTFE architecture

PULVERISETTE 0: Vibratory Micro Mill



SYSTEM:

Vibratory Micro Mill (P0)

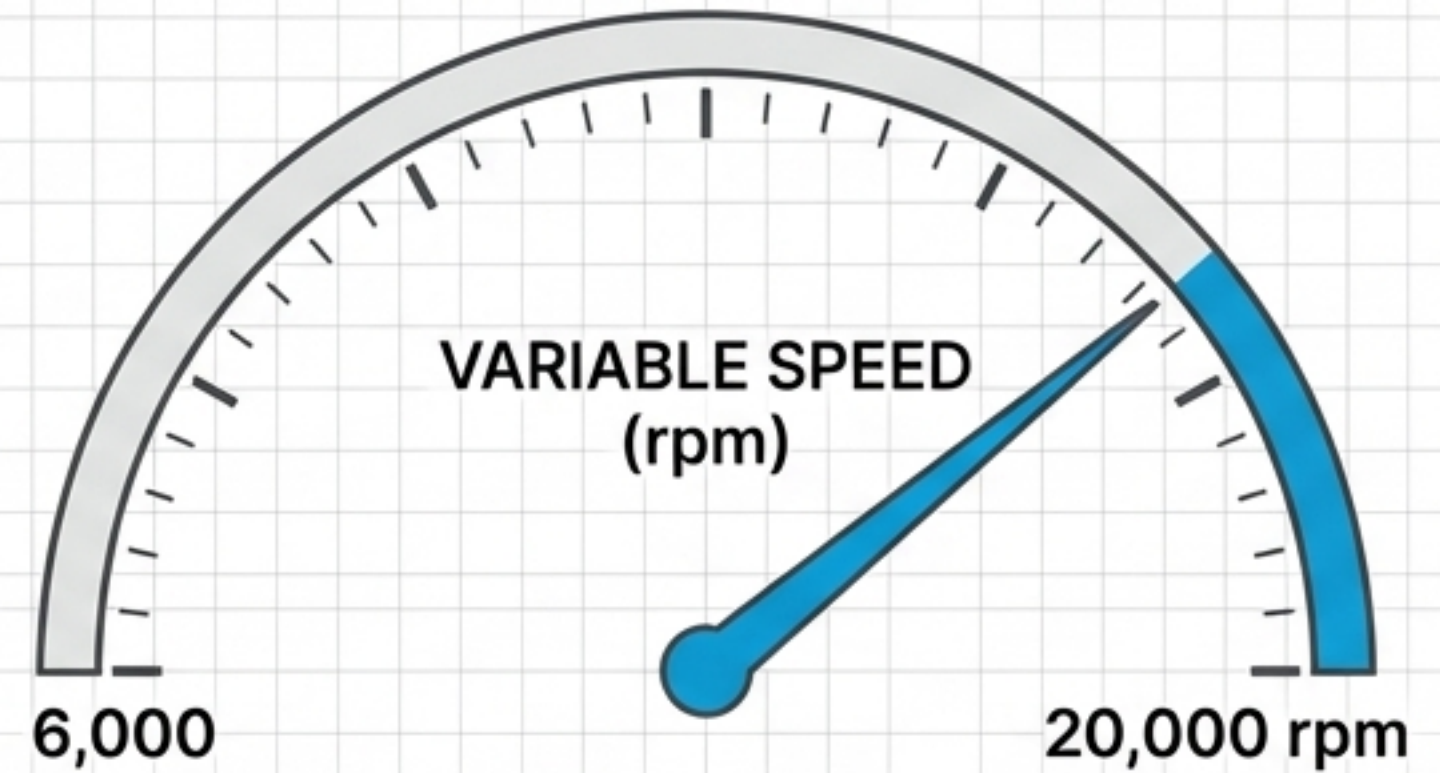
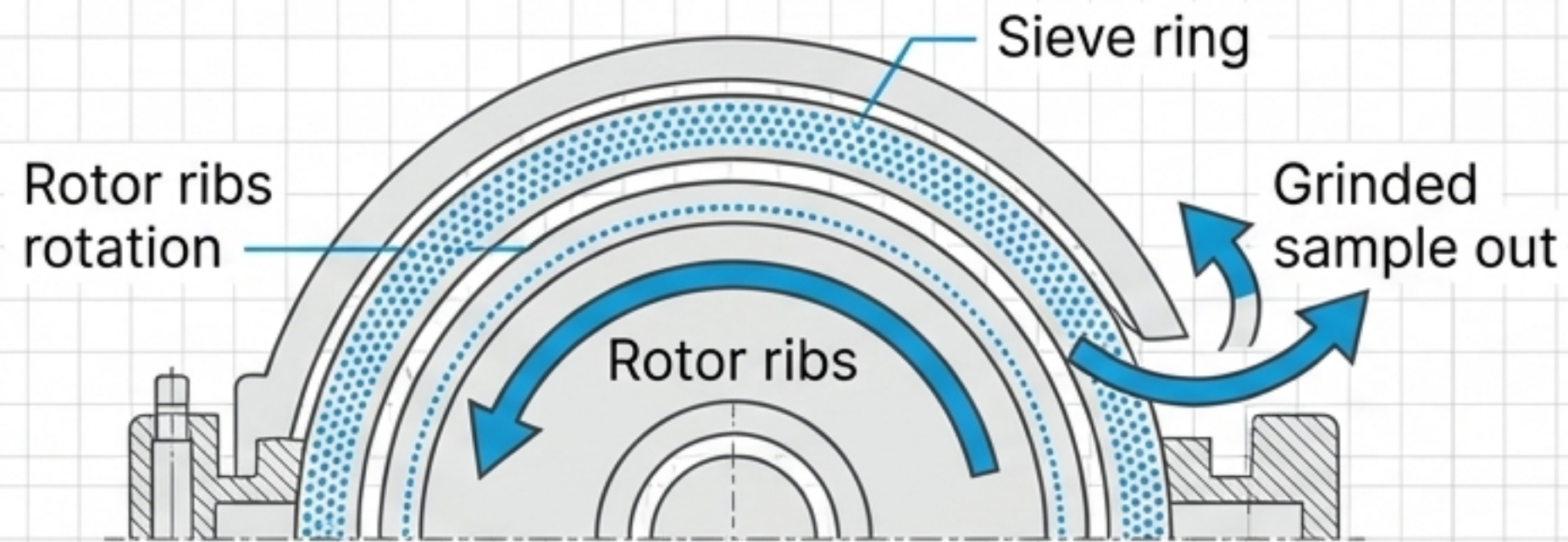
MECHANISM:

Vertical impact vibration

TARGET MATERIALS:

Biological tissues (DNA, Hair, Bone) and RoHS components (Keypads, Glass panels)

Thermal & Variable Speed Milling



- RIB CONFIGURATION: Selectable 8, 12, or 24 ribs.
- OUTPUT CALIBRATION: Sieve Ring tuning from 0.08mm to 6.0mm.
- OPTIMAL SAMPLES: Heat-Sensitive Profiles (PVC, Polymers, Pharmaceuticals, Resins).

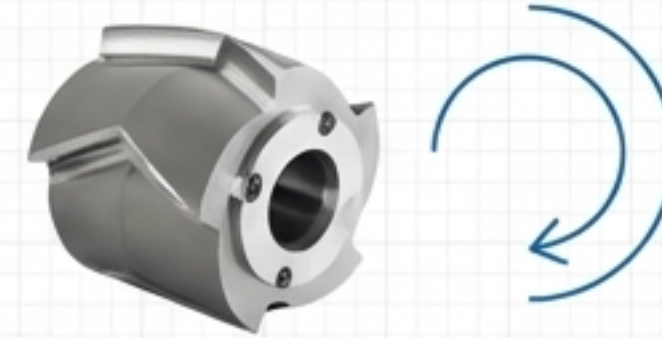
Macro-Shearing & Dissection: Cutting Mills



MAXIMUM INTAKE:

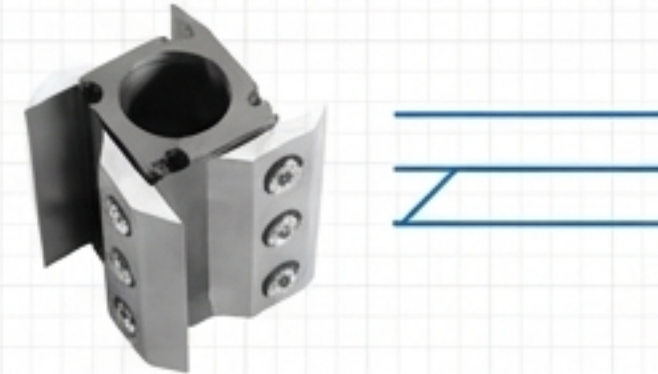
Up to 120 x 85 mm feed size.
Continuous Feeding & Batch Processing.

V-Cut Standard Rotor



Optimized for standard pellets and granules.

Straight Edge Rotor



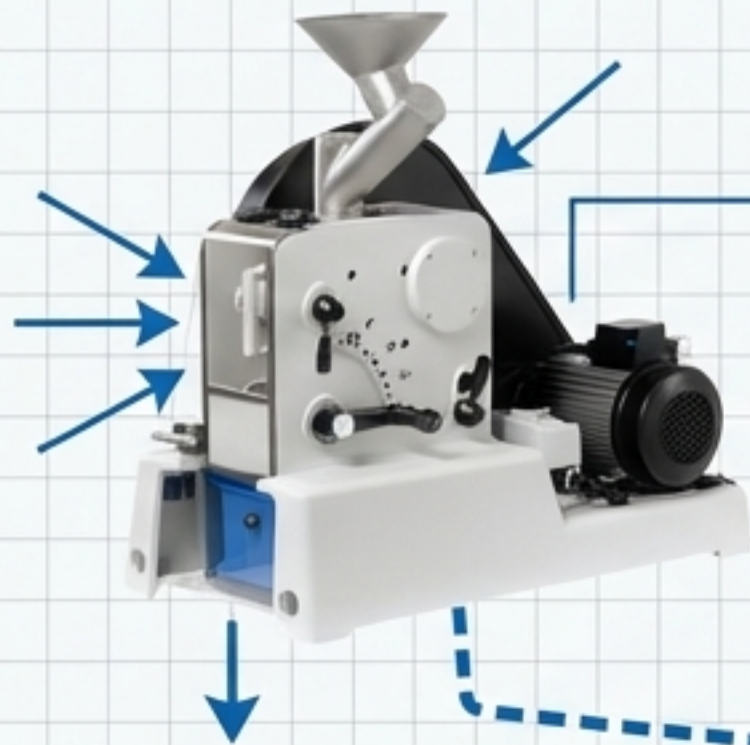
Calibrated for elongated wood and plant fibers.

Disk Milling Cutter



Engineered for heavy boards and electronic waste.

The Rock-to-Sugar Sequence: Pre-Crushing Protocol



PULVERISETTE 1
JAW CRUSHER

01

STAGE 01: JAW COMPRESSION.

Intake: 95mm.
Output: 15mm.



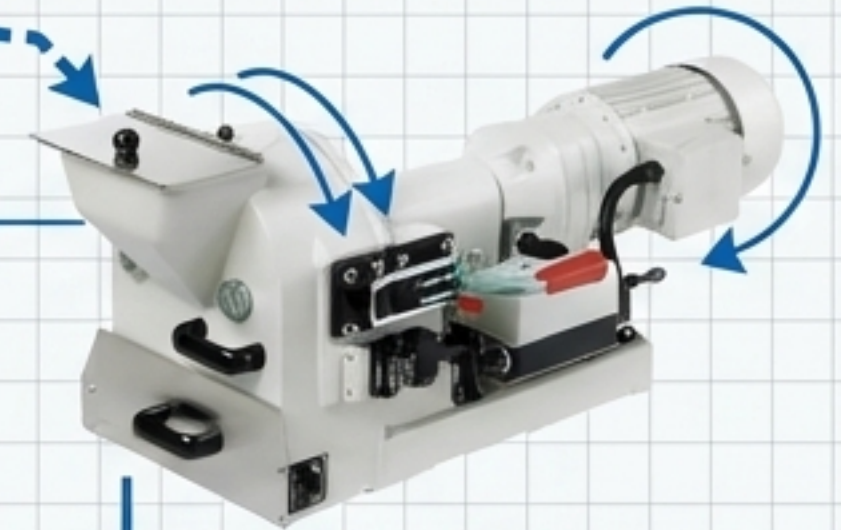
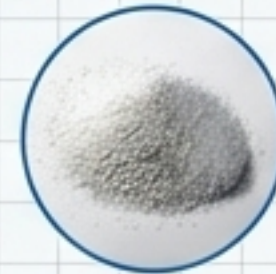
02

CONTINUOUS BATCH TRANSFER.

03

STAGE 02: DISK SHEARING.

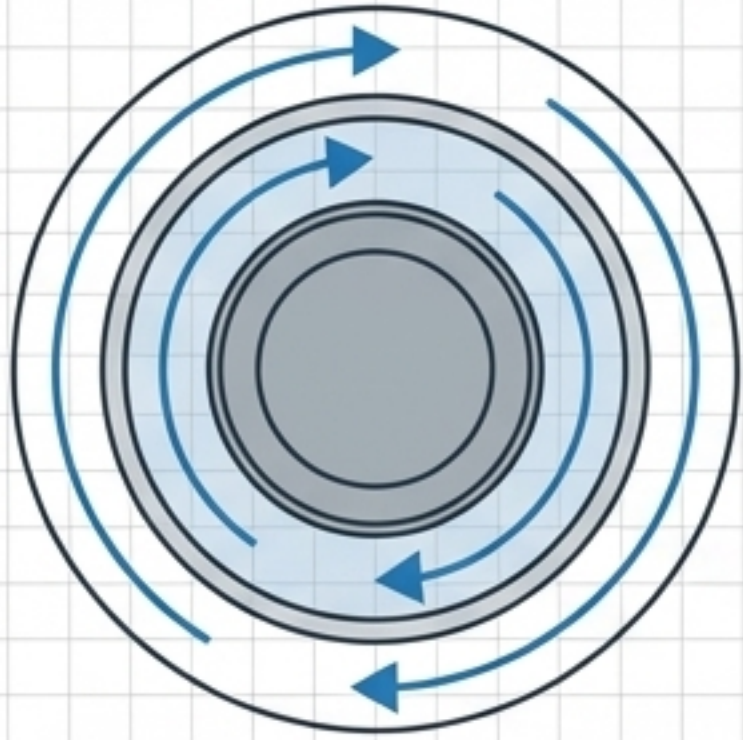
Intake: 20mm.
Output: 0.1mm.



PULVERISETTE 13
DISK MILL

APPLICATIONS: Heavy Geology, Mining
Extraction, Slag, Concrete, Silicates.

Specialized Impact Protocols

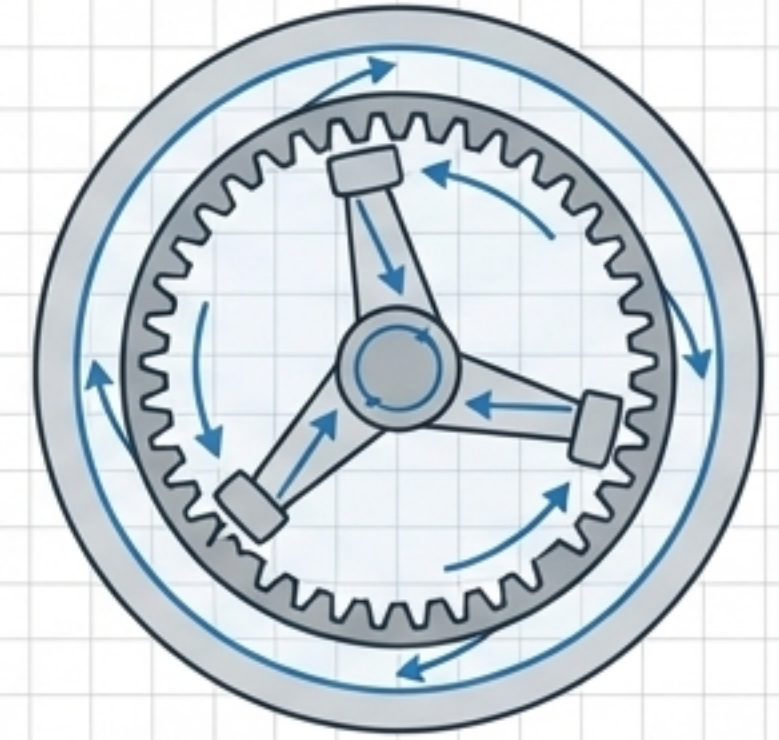


PULVERISETTE 9 (CUP MILL)

DYNAMICS:
Impact & Friction

PERFORMANCE:
12mm input → 20µm output

SECTOR:
XRF Analysis Prep, Metallurgy



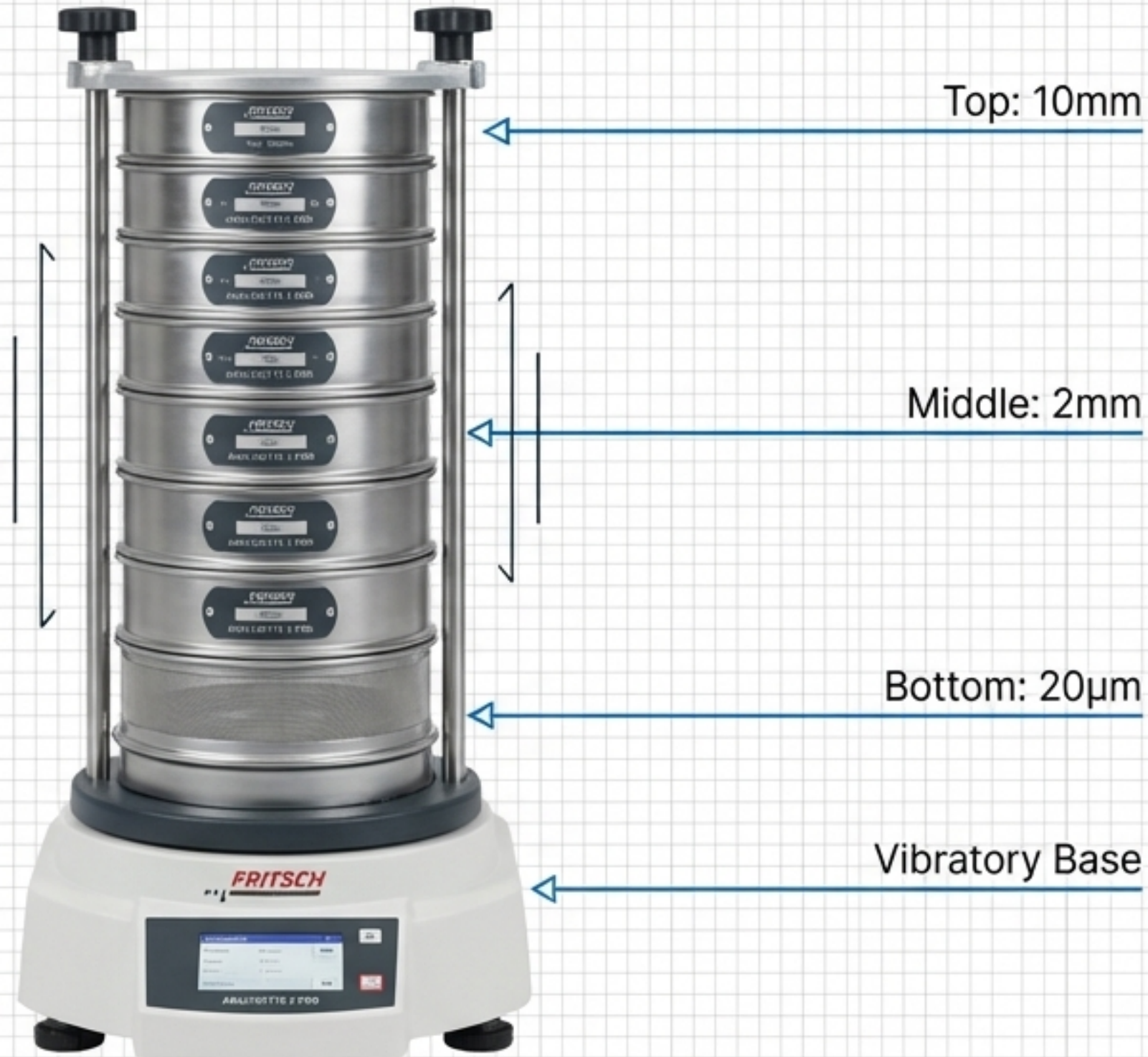
PULVERISETTE 16 (CROSS BEATER)

DYNAMICS: High-speed chamber impacts (2,850 rpm)

PERFORMANCE: Bottom sieve calibration (0.12 - 10mm).
Mohs Hardness limit: 6.

SECTOR: Agriculture, Coal, Minerals

Precision Sieving Telemetry



SYSTEM:

ANALYSETTE 3 PRO & 18

CONTROL SYSTEM:

Electromagnetic Amplitude Control.

CAPABILITY:

ISO 9001 Compliant Wet & Dry Sieving.

RESOLUTION:

Micro-sieving down to 20µm.

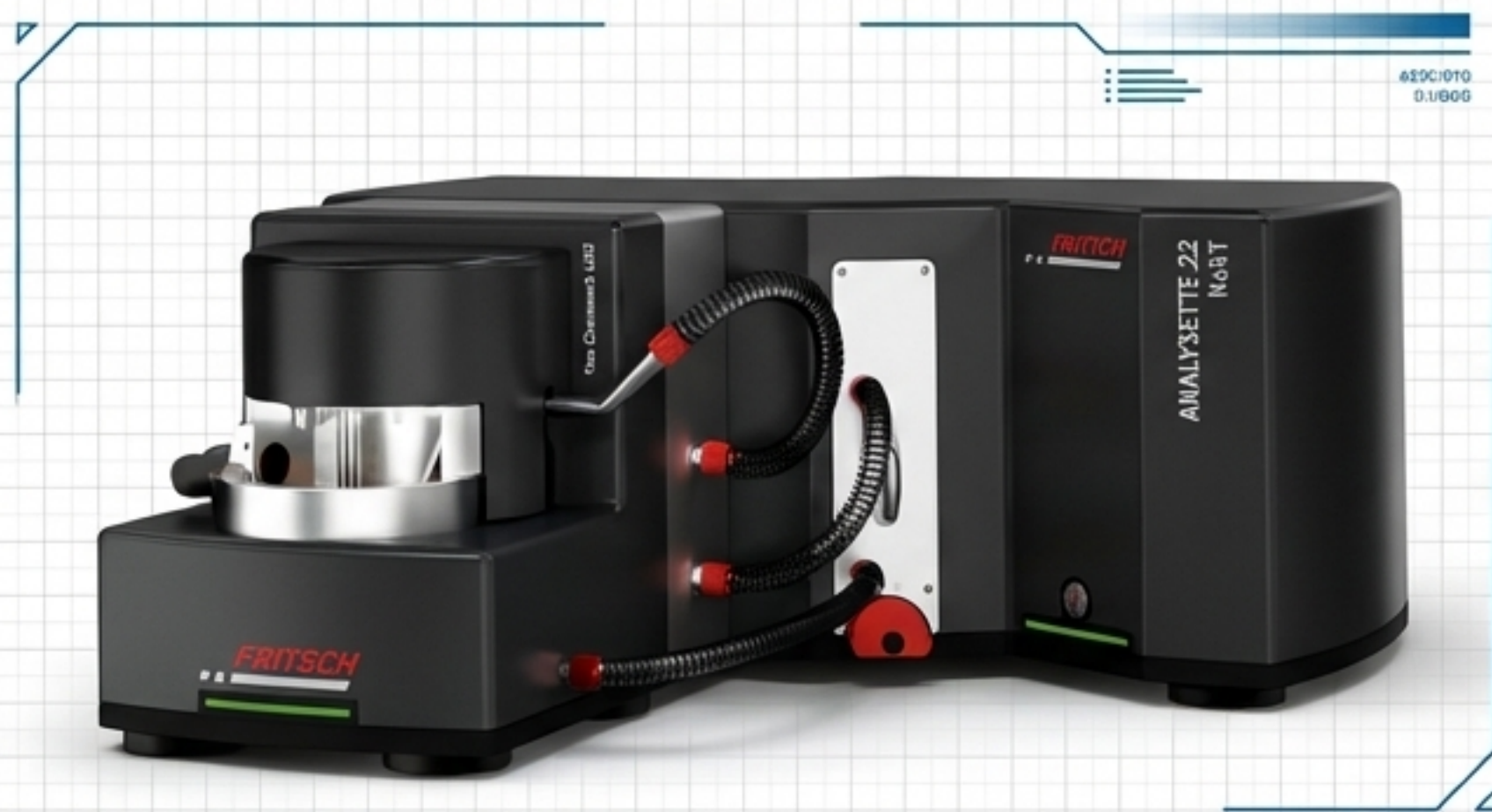
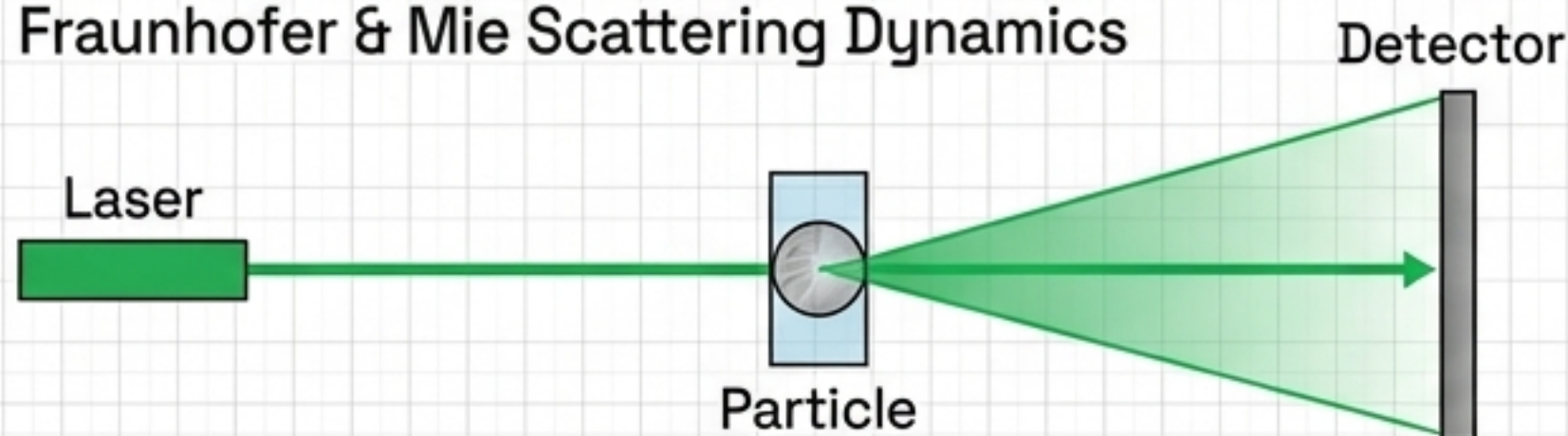


MODULARITY

Converts to Vibratory Micro Mill (PULVERISETTE 0) for <10ml processing via accessory integration.

Laser Diffraction Analysis

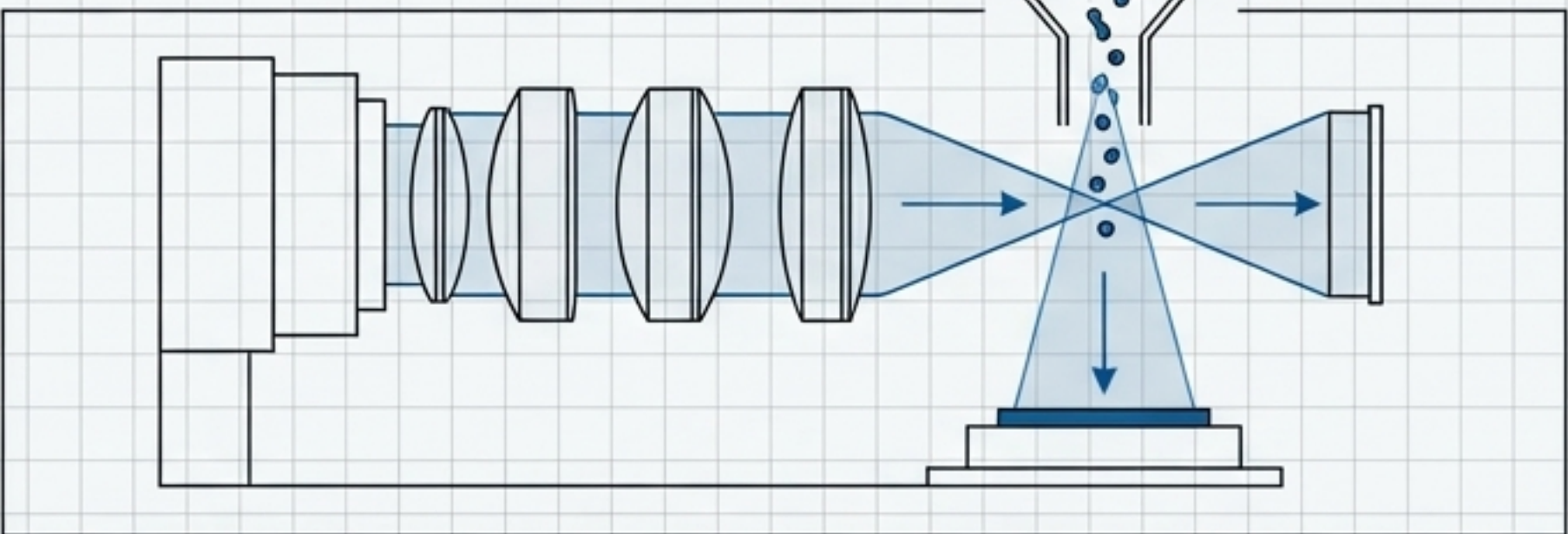
Fraunhofer & Mie Scattering Dynamics



NeXT Micro	NeXT Nano
MEASURING RANGE	
Micro: 0.5 ~ 1,500 μm	Nano: 0.01 ~ 3,800 μm
LARGE ANGLE DETECTOR	
Micro: No	Nano: Yes
BACKWARD SCATTERING	
Micro: No	Nano: Yes
AUTOMATION: Continuous measurement, analysis, rinsing, and SOP reporting.	

Optical Profiling Matrix

Hardware



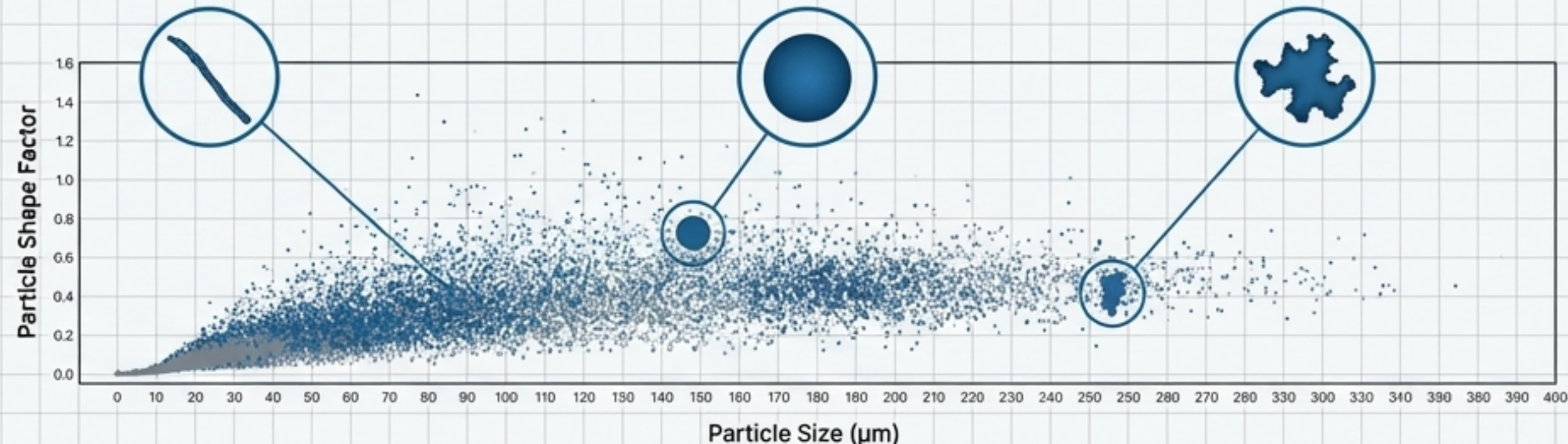
HARDWARE: 5 Megapixel CMOS Particle Profiling with Telecentric Lenses.

Software/Data

FRAME RATE: 75 images/second automatic software analysis (ISS).

MEASUREMENT ENVELOPES:

- Dry Module: 20µm ~ 20mm
- Wet Module: 5µm ~ 3mm



Logistics & Sample Control



100 rpm rotation



LABORETTE 27 (Rotary Cone Divider)

- **PRECISION:** 99.9% Division Accuracy.
- **ROUTING:** Selectable 8, 10, or 30 channel distribution at 100 rpm.



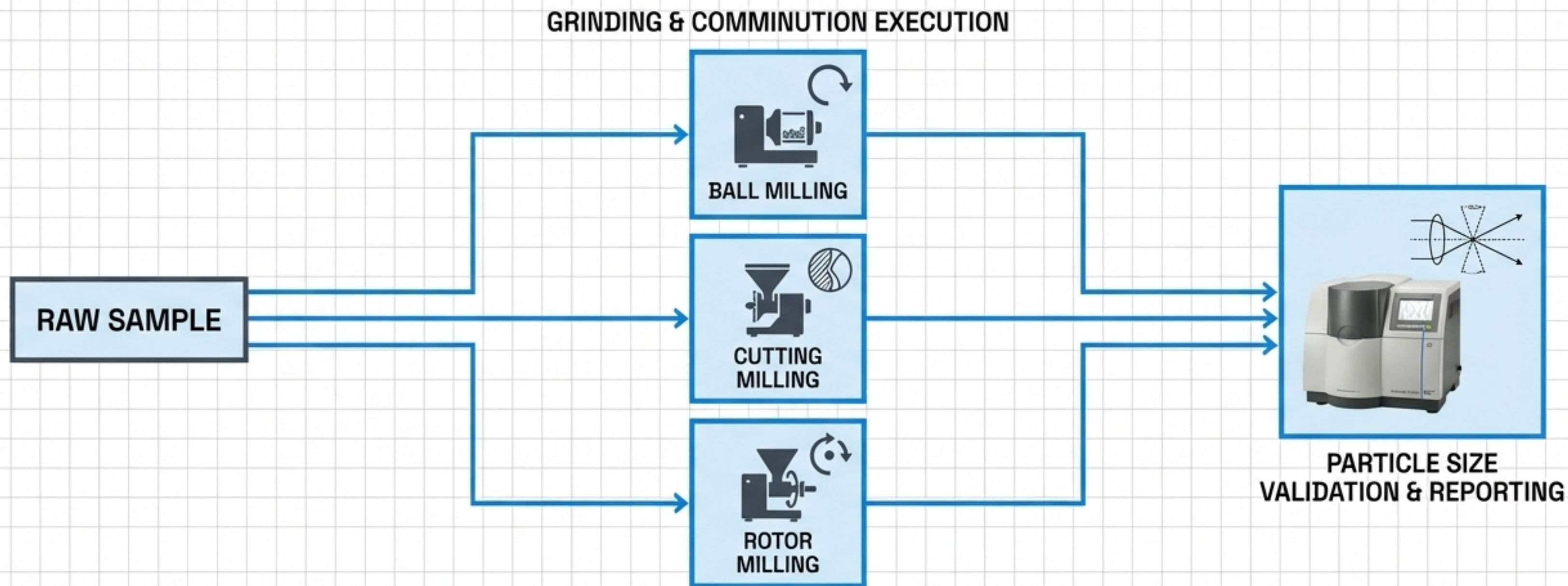
vibratory frequency



LABORETTE 24 (Vibratory Feeder)

- **FEED RATE:** U-Shaped (Up to 2,500g/min) / V-Shaped (Up to 1,500g/min).
- **PURPOSE:** Critical for reproducible, automated laboratory feeding.

The One-Step Ecosystem



INFRASTRUCTURE: Tamyung Science Sample Test Laboratory.

End-to-End Architecture: Optimal Equipment Selection -> Precise Execution -> Validated Results.



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